

8(a)	packet / quantum of <u>energy</u>	M1
	of electromagnetic radiation	A1
8(b)(i)	electron(s)	B1
8(b)(ii)	X labelled – and Y labelled +	B1
8(c)(i)	0.032 MeV	A1
8(c)(ii)	momentum = E / c	C1
	momentum = $(0.032 \times 1.60 \times 10^{-13}) / (3.00 \times 10^8)$ $= 1.7 \times 10^{-23} \text{ N s}$	A1
8(c)(iii)	$E = hf$ and $\lambda = c / f$	C1
	$\lambda = hc / E$ $= (6.63 \times 10^{-34} \times 3.00 \times 10^8) / (0.032 \times 1.60 \times 10^{-13})$	C1
	$\lambda = 3.9 \times 10^{-11} \text{ m}$	A1
8(d)	discussion of bone and soft tissue	B1
	discussion of different attenuation (coefficients) or discussion differences in penetration / transmission / absorption	B1
	<u>transmitted intensities</u> (by bone and tissue) are very different (leading to good contrast images)	B1